Exercises in *Fourier Analysis on Number Fields*

Solutions to the exercises of Ramakrishnan–Valenza

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1 Topological Groups

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Exercise 7.1. Let K be a local field and let ψ be a nontrivial unitary character on K; that is, a continuous, nontrivial homomorphism from K as an additive group into S^1, the circle group. For each $a \in K$, define $\psi_a \colon K \to S^1$ by

$$\psi_a(x) = \psi(ax).$$

(a) Show that ψ_a is again a unitary character of K, and that ψ_a is trivial if and only if $a = 0$.

(b) Show that $a \mapsto \psi_a$ defines a continuous, injective homomorphism α_ψ of K into its Pontryagin dual $\widehat{K}$.

(c) Show that the image of α_ψ is dense in $\widehat{K}$. [*Hint*: Show that $\psi_a(b) = 1$ for all $a \in K$ if and only if $b = 0$.]

(d) Show that α_ψ is bicontinuous.

(e) Show that $\alpha_\psi(K)$ is a complete, hence closed, subgroup of $\widehat{K}$. Conclude that α_ψ is an isomorphism of topological groups.

Solution. (a) Multiplication by a, $x \mapsto ax$, is a continuous endomorphism of the additive group K. Hence ψ_a, its composite with ψ, is a continuous homomorphism $K \to S^1$, i.e. a unitary character of K.

If $a = 0$, then $\psi_a(x) = \psi(0) = 1$ for all $x \in K$, so ψ_a is trivial. Conversely, let $a \neq 0$. Since ψ is nontrivial, there is some $y \in K$ with $\psi(y) \neq 1$, and then $\psi_a(a^{-1}y) = \psi(y) \neq 1$. So ψ_a is nontrivial.

(b) Recall that $\widehat{K}$ has the compact-open topology. A basis of neighborhoods of a character ψ_0 is given by the sets

$$W(\psi_0, C, \epsilon) = \{\psi \in \widehat{K} : |\psi(x) - \psi_0(x)| < \epsilon, \forall x \in C\}.$$

where $C \subset K$ is a compact set and $\epsilon > 0$.

First, reduce to continuity at 0. Note that $\psi_{a+b}(x) = \psi_a(x) \cdot \psi_b(x)$. Then we have that

$$|\psi_a(x) - \psi_b(x)| = |\psi_b| \cdot |\psi_{a-b}(x) - 1| = |\psi_{a-b}(x) - 1|.$$

So it suffices to show that for each compact C and $\epsilon > 0$, there is a $\delta > 0$, such that $\forall |a| < \delta$, $|\psi_a(x) - 1| < \epsilon$.

Given an $\epsilon > 0$. Since ψ is continuous at 0 and $\psi(0) = 1$, then there is an $r > 0$, such that $\forall |x| < r$, $|\psi(x) - 1| < \epsilon$. Since C is compact, then $M = \max_{x \in C}\{|x|\} < \infty$. Choose $\delta = r/M$, then $\forall |a| < \delta$ and $\forall x \in C$, $|ax| < r$. It follows that

$$|\psi_a(x) - 1| = |\psi(ax) - 1| < \epsilon.$$

Injectivity is followed from (a).

(c) Note that

$$\forall a \in K, \psi_a(b) = 1, \Leftrightarrow \forall a \in K, \psi(ab) = 1, \Leftrightarrow b = 0.$$

Recall that we have the fact of Pontryagin dual: let G be a locally compact abelian group and H be a proper closed subgroup of G. Then there is a nontrivial character χ of G, such that $\chi|_H = 1$.

Let H be the closure of the image of α_ψ, i.e. $H = \overline{\alpha_\psi(K)}$. Assume that $H \neq \widehat{K}$. Then we have a character $\chi \in \widehat{\widehat{K}}$ such that χ is trivial on H. And by Pontryagin duality $K \equiv \widehat{\widehat{K}}$, every character of $\widehat{K}$ is evaluation at some point $b \in K$, i.e. $\psi \mapsto \psi(b)$. Then we have that $\chi = (\psi \mapsto \psi(b_0))$ for some $b_0 \in K$. Since χ is trivial on H then $\forall a \in K, \psi_a(b_0) = 1$. It follows that $b_0 = 0$, which contradicts to χ is a nontrivial character of $\widehat{K}$.

□

Exercise 7.2. Let K be a non-Archimedean local field with ring of integers $\mathfrak{o}_K$ and maximal ideal P. Fix a nontrivial unitary character ψ of K. Let P^m be the conductor of ψ; that is, the largest fractional ideal P^m on which ψ is trivial. Let α_ψ, as defined in Exercise 7.1, be the isomorphism between K and its Pontryagin dual.

(a) Show that via α_ψ, P^m identifies with the Pontryagin dual of $\widehat{\mathfrak{o}}_K$. (This is also the inverse different of K, which soon makes its appearance.)

(b) Let $n \geq 1$ and $x \in K$. Show that $xP^n \subseteq \ker(\psi)$ if and only if $x \in P^{m-n}$.

Exercise 7.3. In this exercise we construct nontrivial characters for non-Archimedean local fields, including the standard character for $\mathbb{Q}_p$ and its extensions, and the standard character for the completion of $\mathbb{F}_p(t)$ and its extensions, at the place defined by t^{-1}.

(a) Given any $x \in \mathbb{Q}_p$, let n be the smallest nonnegative integer such that $p^n x$ lies in $\mathbb{Z}_p$. Let $r \in \mathbb{Z}$ be such that $r \equiv p^n x \pmod{p^n}$. Put

$$\psi(x) = e^{2\pi i r/p^n}.$$

Show that $\psi\colon \mathbb{Q}_p \to S^1$ is a nontrivial unitary character of conductor $\mathbb{Z}_p$ and that it identifies with the composite map $\mathbb{Q}_p \to \mathbb{Q}_p/\mathbb{Z}_p \to \mathbb{Q}/\mathbb{Z} \to S^1$.

(b) Let K_∞ denote the completion of $K = \mathbb{F}_p(t)$ defined by the place t^{-1}. Show that $K_\infty = \mathbb{F}_p((t^{-1}))$; that is, that each $x \in K_\infty$ can be uniquely represented by a formal power series

$$\sum_{n=-\infty}^{r} a_n t^n$$

with $a_n \in \mathbb{Z}/p\mathbb{Z}$ for all n less than or equal to the integer r. Put

$$\psi_\infty(x) = e^{2\pi i a_1/p} \in S^1.$$

Show that ψ_∞ is a nontrivial unitary character of K_∞ of conductor $\mathbb{F}_p[[t^{-1}]]$.

(c) Let π be any irreducible polynomial in $\mathbb{F}_p[t] \subseteq K$, and let K_π denote the completion of K defined by the prime ideal (π), with residue field k, which we may identify with a transversal of (π) in $\mathbb{F}_p[t]$. Show that every $x \in K_\pi$ can be written in the form

$$\sum_{n=r}^{+\infty} a_n \pi^n$$

for some integer r, with each coefficient $a_n \in k$. Next put

$$\psi_\pi(x) = e^{2\pi i \operatorname{tr}_{k/\mathbb{F}_p}(a_{-1})/p}.$$

Show that ψ_π is a nontrivial unitary character of K_π.

(d) Let K be any non-Archimedean local field. Show that K is a finite separable extension of a local field K_0 as in parts (a), (b), or (c) above. Let ψ_0 denote the corresponding character of K_0 defined therein. Given $x \in K$, set

$$\psi(x) = \psi_0(\mathrm{tr}_{K/K_0}(x)).$$

Show that ψ is a nontrivial unitary character of K.

(e) Continuing in the same context, show that when the characteristic of K is zero, the conductor of ψ is the inverse different $\mathcal{D}_K^{-1}$. In this case show also that for f the characteristic function of $\mathfrak{o}_K$, the Fourier transform of f is $N(\mathcal{D}_K)^{-1/2}$ times the characteristic function of $\mathcal{D}_K^{-1}$.

Exercise 7.4. We shall now construct the standard character on $\mathbb{A}_K/K$ for an algebraic number field K.

(a) For each rational prime p, let ψ_p denote the standard character of $\mathbb{Q}_p$ defined above. Also define $\psi_\infty \colon \mathbb{R} \to S^1$ by $\psi_\infty(x) = e^{-2\pi i x}$. Show that

$$\psi \colon \mathbb{A}_\mathbb{Q} \to S^1, \qquad (x_v) \mapsto \prod_v \psi_v(x_v)$$

is a nontrivial unitary character on $\mathbb{A}_\mathbb{Q}$, which is trivial on $\mathbb{Q}$ (embedded diagonally). [*Hint*: Use the Artin product formula.]

(b) For any number field K, define $\psi_K \colon \mathbb{A}_K \to S^1$ by

$$\psi_K(x) = \prod_v \psi_v(x_v)$$

where v ranges over all the places of K and ψ_v is the standard character on the local field K_v, as constructed above. Show that ψ_K is a nontrivial unitary character on $\mathbb{A}_K$.

(c) Show also that $\psi_K(x) = \psi(\mathrm{tr}(x))$ where tr denotes the adelic trace map from K to $\mathbb{Q}$. Conclude from part (a) that ψ_K is trivial on K.

Exercise 7.5. We shall now construct the standard character for the completion K_v of $K = \mathbb{F}_p(t)$ at any *finite* place v.

(a) Let $\pi(t)$ be an irreducible element of $\mathbb{F}_p[t]$ of degree $d \geq 1$. Show that K has a unique place v such that the polynomial $\pi(t)$ generates the maximal ideal of $\mathfrak{o}_v = \{\alpha \in K_v : |\alpha|_v \leq 1\}$. Show further that $q_v = \mathrm{Card}(\mathfrak{o}_v/\pi\mathfrak{o}_v) = p^d$, and that the polynomials in $\mathbb{F}_p[t]$ of degree less than d in fact generate $\mathfrak{o}_v/\pi\mathfrak{o}_v$.

(b) Given any place v of K (possibly infinite), let us define

$$K^{(v)} = \{x \in K : |x|_u \leq 1 \ \forall u \neq v\}.$$

Show that $K_v = K^{(v)} \oplus \mathfrak{o}_v$. [*Hint*: By replacing t by t^{-1} if necessary, one may assume that v is defined by an irreducible polynomial $\pi(t) \in \mathbb{F}_p[t]$ and then apply the previous part.]

(c) Let v be any finite place of K; that is, one defined by an irreducible polynomial $\pi(t) \in \mathbb{F}_p[t]$ of positive degree d. Show that every element y of $K^{(v)}$ can be written as z/π^n for some $n \geq 0$ and polynomial $z = z(t) \in \mathbb{F}_p[t]$ of the form

$$c_1 t^{nd-1} + c_2 t^{nd-2} + \cdots + c_{nd}$$

with $c_j \in \mathbb{F}_p$ for all j, possibly all zero. If $x = y + \alpha \in K_v$ with $\alpha \in \mathfrak{o}_v$, put

$$\psi_v(x) = e^{2\pi i c_1/p}.$$

(This is of course well-defined despite the status of c_1 as a residue class modulo the prime p.) Show that ψ_v is a nontrivial character of K_v of conductor $\mathfrak{o}_v$.

(d) Let v, π, and y be as in the previous part. Show that $y \equiv c_1 t^{-1} \pmod{t^{-2}}$. Conclude that $\psi_v(y)\psi_\infty(y) = 1$, where ψ_∞ is as in part (b) of Exercise 7.3. [*Hint*: Write π as $t^d g$, with g a polynomial in t^{-1} of constant term 1, so that $y = zg^{-n}t^{-nd}$.]

Exercise 7.6. We may now use the results of previous problems to construct the standard character $\psi_K \colon \mathbb{A}_K \to S^1$ for K any global field of positive characteristic.

(a) Suppose that $K_0 = \mathbb{F}_p(t)$. Define ψ_{K_0} by

$$\psi_{K_0}(x) = \prod_v \psi_v(x_v)$$

where ψ_∞ is given by Exercise 7.3 and ψ_v (v finite) is given by part (c) of the previous exercise. Show that ψ_{K_0} is a nontrivial character of $\mathbb{A}_{K_0}$ that, moreover, is trivial on K_0.

(b) Now let K be any finite separable extension of K_0. Use the trace map from K to K_0 (or, equivalently, the corresponding adelic trace map) to define the standard character ψ_K on K. Show that this character is nontrivial on $\mathbb{A}_K$, but trivial on K.

Exercise 7.7. Let G be a locally compact abelian group with Haar measure $\mathrm{d}g$, and let $\mathrm{d}\widehat{g}$ be the dual measure; that is, the measure on the Pontryagin dual $\widehat{G}$ relative to which the Fourier inversion formula holds. Suppose that we have an isomorphism $\alpha \colon G \xrightarrow{\cong} \widehat{G}$ of topological groups.

(a) Show that there is a *unique* multiple $\mu = t \cdot \mathrm{d}g$ such that μ identifies with its dual measure under α. One calls μ the *self-dual* measure on G relative to the isomorphism α.

(b) Let G be the additive group of a non-Archimedean local field K, $\psi \colon K \to S^1$ the standard unitary character, and α_ψ the isomorphism of Exercise 7.1. Denote by $\mathrm{d}x$ the self-dual measure relative to ψ (that is, relative to α_ψ). Show that $\mathrm{vol}(\mathfrak{o}_K, \mathrm{d}x) = N(\mathcal{D}_K)^{-1/2}$, where $\mathcal{D}_K$ denotes the different of K.

Exercise 7.8. In this exercise we analyze the dependence of the epsilon factor on the additive character ψ and the Haar measure $\mathrm{d}x$. Let K be a local field and consider $\varepsilon(\chi, \psi, \mathrm{d}x)$ for any $\chi \in \mathrm{Hom}_{\mathrm{cont}}(K^\times, \mathbb{C}^\times)$.

(a) For every positive real number t, show that

$$\varepsilon(\chi, \psi, t \cdot \mathrm{d}x) = t \cdot \varepsilon(\chi, \psi, \mathrm{d}x).$$

(b) Let $a \in K^\times$, and let ψ_a denote the character defined by $\psi_a(x) = \psi(ax)$. Show that

$$\varepsilon(\chi, \psi_a, \mathrm{d}x) = \chi(a)|a|^{-1}\varepsilon(\chi, \psi, \mathrm{d}x).$$

(c) Let K be non-Archimedean with unique prime ideal P, and let P^n and P^m be the conductors of χ and ψ, respectively. Then show for every unramified character ν of $K^\times$ that

$$\varepsilon(\chi\nu, \psi, \mathrm{d}x) = \nu(\pi^{m+n})\varepsilon(\chi, \psi, \mathrm{d}x)$$

where π is a uniformizing parameter for $\mathfrak{o}_K$. Note that the first factor on the right is well-defined because ν is unramified. [*Hint*: Use the explicit formula given in [RV99, Eq. (7.6)].]

Exercise 7.9. Let K be a local field with standard character ψ and self-dual measure $\mathrm{d}x$, and let $\chi = \omega|\cdot|^s$ be a (quasi-)character on $K^\times$ with ω unitary. Put

$$W(\omega) = \varepsilon(\chi, \psi, \mathrm{d}x)\big|_{s=1/2}.$$

(a) By using the functional equation, show that

$$W(\omega)W(\overline{\omega}) = \omega(-1).$$

Conclude that $|W(\omega)| = 1$. [*Hint*: First prove that $W(\overline{\omega}) = \omega(-1)\overline{W(\omega)}$.]

(b) Using [RV99, Section 7.1], show that for $K = \mathbb{R}$, $W(\omega) = 1$ (respectively, $-i$) for $\omega = 1$ (respectively, $\omega = \mathrm{sgn}$). Show further that when $K = \mathbb{C}$, we have $W(\omega) = 1$.

(c) Let K be non-Archimedean with uniformizing parameter π, and let ω as above have conductor $\pi^n\mathfrak{o}_K$. Put

$$G = \sum_{x \in U/U_n} \overline{\omega}(x)\psi(x\pi^{-d-n})$$

where $\pi^d\mathfrak{o}_K = \mathcal{D}_K$ (the different), $U_r = 1 + \pi^r\mathfrak{o}_K$ ($r > 0$), and U is the full group of units in $\mathfrak{o}_K$. Show directly that $|G|^2 = q^n$. Then appeal to [RV99, Eq. (7.6)] to deduce (once again) that $|W(\omega)| = 1$.

Exercise 7.10. (Tate) Let K be a non-Archimedean local field with uniformizing parameter π, and let ω be a unitary character of $K^\times$ with conductor $\pi^n\mathfrak{o}_K$. Let $\mathfrak{a}$ be an ideal of $\mathfrak{o}_K$ such that $\mathfrak{a}^2$ divides $\pi^n\mathfrak{o}_K$. Put $\mathfrak{b} = \mathfrak{a}^{-1}\pi^n\mathfrak{o}_K \subseteq \mathfrak{o}_K$.

(a) Show that there exists an element $c \in K$ such that $c\mathfrak{o}_K = \pi^n\mathcal{D}_K$ and also

$$\omega(1 + t) = \psi(c^{-1}t)$$

for all $t \in \mathfrak{b}$. [*Hint*: Suppose that $\mathfrak{a} \neq \mathfrak{o}_K$, so that π divides $\mathfrak{a}$. If $t, z \in \mathfrak{b}$, then $tz \in \pi^n\mathfrak{o}_K$. Consequently, $\omega(1 + t)\omega(1 + z) = \omega(1 + t + z)$, and therefore the map that sends t to $\omega(1+t)$ is a character of the additive group $\mathfrak{b}$, and this extends to one on K. Now appeal to the isomorphism of K with its dual.]

(b) For c as in the previous part and K now assumed to have characteristic zero, show that

$$W(\omega) = N(\mathfrak{b}\mathfrak{a}^{-1})^{-1/2} \sum_{x \in (1+\mathfrak{a})/(1+\mathfrak{b})} \overline{\omega}(c^{-1}x)\psi(c^{-1}x).$$

NOTE. When $\mathfrak{a} = \mathfrak{o}_K$, this result is identical to [RV99, Eq. (7.6)].

(c) For ω unramified, show that $W(\omega) = \omega(\pi)^d$ if $\pi^d\mathfrak{o}_K = \mathcal{D}_K$.

(d) Let E be a number field containing $W(\omega)$. Then show that for every place v of E not dividing the residual characteristic p of K, we have $|W(\omega)|_v = 1$. Using the Artin product formula, conclude that if E has a unique place u above p, then also $|W(\omega)|_u = 1$.

(e) Suppose that $\pi^n\mathfrak{o}_K = \pi\mathfrak{a}$. Then, using part (b), show that $z = (\omega(c^{-1})W(\omega))^2$ lies in the cyclotomic extension $E = \mathbb{Q}(e^{2\pi i/p^r})$ for some $r \geq 1$. Show further, using part (d), that z must be a root of unity.

(f) (Lamprecht, Dwork) Suppose that ω is either *unramified* (that is, $n = 0$) or *wildly ramified* (that is, $n \geq 2$). Show that $W(\omega)$ is a root of unity. [*Hint*: This is clear if n is even. When n is odd, write $\pi^n\mathfrak{o}_K = \pi\mathfrak{a}$ and apply the previous part.]

Exercise 7.11. Let K be a global field. For any idele class character $\chi = (\chi_v)$, put

$$W(\chi) = \prod_v W(\chi_v).$$

(a) Show that $W(\chi)W(\chi^{-1}) = 1$. In particular, if χ is unitary, then $|W(\chi)| = 1$.

(b) Let K be a number field, and let χ be quadratic or trivial. Conclude from part (a) that $W(\chi) = \pm 1$. [This number is called the *sign of the functional equation* of $L(s, \chi)$.] Show, moreover, that if χ is unramified everywhere, then

$$W(\chi) = \widetilde{\chi}(\mathcal{D}_K)$$

where $\widetilde{\chi}$ is the associated character of the class group Cl_K (cf. [RV99, Proposition 5-19]). [*Hint*: Use the previous problem.]

(c) Let χ be a quadratic idele class character, and let E/K be the quadratic extension corresponding to the open subgroup $\ker(\chi)$ of C_K by class field theory ([RV99, Theorem 6-6]). Note that $\zeta_E(s) = \zeta_K(s)L(s, \chi)$ and deduce the formula

$$\varepsilon(s, \chi) = \left(|d_K|/|d_E|^{1/2}\right)^{1-2s}$$

where d_K and d_E are the discriminants of K and E, respectively.

(d) (Hecke's Theorem, Serre's Proof) Let K be a number field. Then prove:

THEOREM (Hecke). *The ideal class of $\mathcal{D}_K$ is a square in the class group Cl_K.*

[*Hint*: Observe that it suffices to show that $\widetilde{\chi}(\mathcal{D}_K) = 1$ for every quadratic character $\widetilde{\chi}$ of Cl_K. Then appeal to parts (b) and (c) of this problem.]

Exercise 7.12. We consider here the Fourier transforms of Schwartz–Bruhat functions.

(a) Let K be a local field. Show that for every $f \in \mathcal{S}(K)$, its Fourier transform likewise lies in $\mathcal{S}(K)$. [*Hint*: For K Archimedean, this is a well-known classical fact. In the non-Archimedean case, use that f is a linear combination of characteristic functions of the basic compact sets P^n, where P is the unique prime associated with K.]

(b) Let K be a global field and assume that $f \in \mathcal{S}(\mathbb{A}_K)$. Show that the Fourier transform of f is likewise in $\mathcal{S}(\mathbb{A}_K)$. [*Hint*: First prove that $\mathcal{S}(\mathbb{A}_K)$ is generated by factorizable functions $\bigotimes_v f_v$, where each f_v lies in $\mathcal{S}(K_v)$ for all v, and then use the previous part to establish the result in this special case.]

(c) Use the results of [RV99, Chapter 3] to show that the Fourier transform map $\mathcal{S}(\mathbb{A}_K) \to \mathcal{S}(\mathbb{A}_K)$ extends to an isometry $L^2(\mathbb{A}_K) \to L^2(\mathbb{A}_K)$.

Exercise 7.13. (Hecke Characters) Let K be a number field of degree d, let S be a finite set of places containing S_∞, and let $J_K(S)$ be the group of fractional ideals prime to S. Furthermore, let $\mathbb{I}_K(S)$ denote the subgroup of $\mathbb{I}_K$ consisting of ideles $y = (y_v)$ such that $y_v = 1$ at every place v in S.

(a) Define $\alpha\colon \mathbb{I}_K(S) \to J_K(S)$ by

$$\alpha(y) = \prod_{v \notin S} P_v^{v(y_v)}.$$

Show that α is a surjective homomorphism with kernel

$$U(S) = \{y \in \mathbb{I}_K(S) : y_v \in \mathfrak{o}_v^\times \ \forall v \text{ finite}\}.$$

Conclude that every character χ of $\mathbb{I}_K$ that is unramified outside S defines a character $\widetilde{\chi}$ of $J_K(S)$.

(b) Let β be a *Grössencharakter* (or *Hecke character*) of K, which is to say a homomorphism from $J_K(S)$ to $\mathbb{C}^\times$ for which there exists an integral ideal M with support S, complex numbers $s_1, \ldots, s_d$ and integers $m_1, \ldots, m_d$ such that for every $\alpha \in K^\times(M)$, one has

$$\beta((\alpha)) = \prod_{j=1}^{d} \sigma_j(\alpha)^{m_j} |\sigma_j(\alpha)|^{s_j}$$

where $\{\sigma_j\}$ is the set of embeddings of K in $\mathbb{C}$. Show that every β is of the form $\widetilde{\chi}$ for some character $\chi\colon \mathbb{I}_K \to \mathbb{C}^\times$ that is trivial on $K^\times$. [*Hint*: Show that β defines a character of the ray class group $\mathrm{Cl}_K(M)$ and then lift it to the idele class group C_K.]

(c) Let $\Phi = \widetilde{\chi}$ for an idele class character χ unramified outside S. Show that Φ is in fact a Grössencharakter. [*Hint*: Take

$$M = \prod_{v \in S} P_v^{\max(1, n_v)}$$

where n_v is the exponent of the conductor of χ_v. For $\alpha \in K^\times(M)$, show that $\Phi((\alpha)) = \prod_{v \notin S} \chi_v(\alpha) = 1/(\chi_\infty \circ \sigma)(\alpha)$ by using first that $|\alpha|_v = 1$ for all v in S and second that χ is trivial on $K^\times$.]

Exercise 7.14. Fix an integer $k \geq 1$, and consider the *polylogarithm* function

$$l_k(z) = \sum_{n \geq 1} \frac{z^n}{n^k}.$$

Note that $l_1(z) = -\log(1-z)$. The special case of $l_2(z)$ is called *Euler's dilogarithm* function.

(a) Show that the series is normally convergent in the open unit disk $\{|z| < 1\}$ and that it has finite limit on the unit circle S^1 (respectively, $S^1 \smallsetminus \{1\}$) if $k > 1$ (respectively, $k = 1$).

(b) Let χ be a Dirichlet character of conductor $m \geq 1$. For w an mth root of unity, put

$$G(\chi, w) = \sum_{c \bmod m} \chi(c) w^c.$$

Let $L(s, \chi)$ be the Dirichlet L-series, and assume that χ is not the trivial character if $k = 1$. Put $\chi(-1) = (-1)^r$. Then show that

$$L(k, \chi) = \frac{1}{m} \sum_{1 \leq b \leq m/2} G(\chi, e^{-2\pi i b/m}) [l_k(e^{2\pi i b/m}) + (-1)^r l_k(e^{-2\pi i b/m})].$$

[*Hint*: Expand χ in terms of the basis $\{e^{2\pi i b/m} : 0 \leq b < m\}$ of $\mathbb{C}[\mathbb{Z}/m\mathbb{Z}]$.]

(c) Show that

$$G(\chi, e^{-2\pi i b/m}) = \overline{\chi}(b) G(\chi, e^{-2\pi i/m}).$$

Simplify part (b) accordingly for $k = 1$ and deduce [RV99, Proposition 7-26].

(d) For any $k \geq 1$, show that there is an elementary expression for the Dirichlet series $L(k, \chi)$ if $\chi(-1) = (-1)^k$.

Exercise 7.15. Let χ be a Dirichlet character of conductor $m \geq 1$. As above, put

$$W(\chi) = \varepsilon(\chi |\cdot|^{1/2}) \quad (= \varepsilon(\tfrac{1}{2}, \chi)).$$

Show that

$$W(\chi) = \frac{G(\chi)}{\sqrt{m}} (-i)^r$$

where $r = 0$ (respectively, 1) if $\chi(-1) = 1$ (respectively, -1), and

$$G(\chi) = \sum_{a \bmod m} \chi(a) e^{2\pi i a/m}.$$

Exercise 7.16. Let K be a number field and let S be a set of primes in $\mathfrak{o}_K$. One says that S has *lower Dirichlet density*

$$\underline{\delta}(S) = \liminf_{s \to 1^+} \frac{1}{\log \frac{1}{s-1}} \cdot \sum_{P \in S} \frac{1}{N(P)^s}$$

provided that the indicated lim inf exists. The *upper Dirichlet density* $\overline{\delta}(S)$ is likewise defined via the lim sup.

(a) Show that every S has both a lower and an upper Dirichlet density. [*Hint*: Use [RV99, Proposition 7-31], part (ii), and also that a bounded sequence in $\mathbb{R}$ has an infimum and a supremum.]

(b) Show that S has a Dirichlet density (as defined previously) if and only if $\underline{\delta}(S) = \overline{\delta}(S)$.

(c) Let T be any finite set of primes in $\mathfrak{o}_K$. Show that $\underline{\delta}(S \cup T) = \underline{\delta}(S)$ and $\overline{\delta}(S \cup T) = \overline{\delta}(S)$. Conclude, in particular, that any finite set of primes has density zero.

(d) Let $\mathscr{P}_K^1$ denote the set of degree-one primes of K; that is, prime ideals P of $\mathfrak{o}_K$ such that $N(P)$ is a rational prime. Show that for any set S of primes, $\underline{\delta}(S \cap \mathscr{P}_K^1) = \underline{\delta}(S)$ and $\overline{\delta}(S \cap \mathscr{P}_K^1) = \overline{\delta}(S)$. Conclude that the set of primes in $\mathfrak{o}_K$ of degree greater than one has density zero.

Exercise 7.17. Let E/K be a finite abelian extension of number fields with Galois group $G = \mathrm{Gal}(E/K)$. Let $U = N_{E/K}(C_E)$. Recall that C_K/U is isomorphic to G via class field theory. (See [RV99, Theorem 6-6].) For any character χ of G, let $\widetilde{\chi}$ denote the corresponding character of C_K that is trivial on U. Keeping in mind that the extension E/K is abelian, for any prime P of K, we let

$$\varphi_P = \left(\frac{E/K}{P}\right)$$

denote the corresponding Frobenius element. Now set

$$\chi(P) = \begin{cases} \chi(\varphi_P) & \text{if } P \text{ is unramified in } E \text{ or if } \chi \text{ is trivial,} \\ 0 & \text{otherwise,} \end{cases}$$

and put

$$L(s, \chi) = \prod_P (1 - \chi(P) N(P)^{-s})^{-1}.$$

(a) Show that $L(s, \chi)$ converges absolutely in $\{\mathrm{Re}(s) > 1\}$ and further admits a meromorphic continuation to the whole s-plane, with no poles except possibly a simple one at $s = 1$, which occurs only when $\chi = 1$. Show, moreover, that there is a functional equation relating s to $1 - s$. [*Hint*: Use [RV99, Theorems 6-6 and 7-16].]

(b) Show that

$$\zeta_E(s) = \prod_{\chi \in \widehat{G}} L(s, \chi).$$

[*Hint*: Write the left-hand side as an Euler product over the primes P of K.]

(c) Show that

$$\lim_{s \to 1^+} \frac{1}{\log \frac{1}{s-1}} \sum_{P \in \mathscr{P}_K^1} \frac{\chi(P)}{N(P)^s} = \begin{cases} 1 & \text{if } \chi = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The following four problems lead the reader through a proof of a version of the Tchebotarev density theorem, here reformulated in terms of Dirichlet density. Recall that $\mathscr{P}_{K,f}$ denotes the set of finite places of K, or, equivalently, the set of prime ideals of $\mathfrak{o}_K$.

THEOREM. *Let E/K be a finite Galois extension of number fields with Galois group G, and let C be a conjugacy class in G. Set*

$$S_K(C) = \Big\{P \in \mathscr{P}_{K,f} : \left(\frac{E/K}{P}\right) = C\Big\}.$$

Then $\delta(S_K(C))$ exists and equals $\operatorname{Card}(C)/\operatorname{Card}(G)$.

We shall also need the degree-one version of $S_K(C)$, which we define as

$$S_K^1(C) = S_K(C) \cap \mathscr{P}_K^1.$$

The first problem is well known and deals with the abelian case. The next three encapsulate the ideas from an elegant proof due to O. Schreier.

Exercise 7.18. Let E/K be a finite abelian extension, so that C reduces to a singleton set $\{a\}$ for some $a \in G$. We recall from [RV99, Chapter 3, Exercise 13], the discrete form of the Fourier inversion formula: if f is a complex-valued function on G, then

$$f(g) = \frac{1}{\operatorname{Card}(G)} \sum_{\chi \in \widehat{G}} \widehat{f}(\chi)\chi(g)$$

where

$$\widehat{f}(\chi) = \sum_{g \in G} f(g)\overline{\chi}(g).$$

(a) Given $a \in G$, define $f_a \colon G \to \mathbb{C}$ by

$$f_a(g) = \sum_{\chi \in \widehat{G}} \chi(a^{-1}g).$$

Show that f_a is in fact the characteristic function of $\{a\}$.

(b) For all $g \in G$ show that

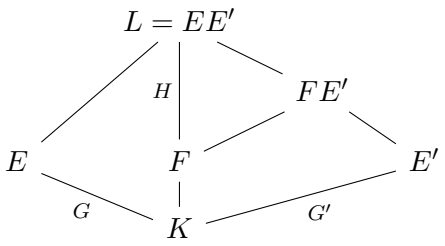
$$\sum_{P \in S_K(C)} \frac{1}{N(P)^s} = \sum_{g \in G} \sum_{P \in S_K(C)} \frac{f_a(g)}{N(P)^s} = \frac{1}{\operatorname{Card}(G)} \sum_{\chi \in \widehat{G}} \sum_{P} \frac{\chi(P)}{N(P)^s}.$$

(c) Show that $\delta(S_K(C))$ exists and equals $1/\operatorname{Card}(G) = \operatorname{Card}(C)/\operatorname{Card}(G)$. [*Hint*: Use the previous exercise.]

Exercise 7.19. Suppose that E/K is any finite normal extension of number fields. Let G be the associated Galois group, and let C be a conjugacy class in G. Suppose further that E' is an auxiliary finite abelian extension of K with Galois group G' such that E and E' are linearly disjoint over K. Put $L = EE'$.

(a) Show that L/K is normal with Galois group $\mathcal{G} = G \times G'$.

(b) Choose any conjugacy class $C' = \{a\}$ in G', and let H be the subgroup generated by CC' in $\mathcal{G}$. Put $F = L^H$, and consider the compositum FE'. We have the following diagram:



Show that FE' is a normal extension of F with Galois group given by

$$\mathrm{Gal}(FE'/F) \cong H/(H \cap \langle C \rangle) \cong \langle C' \rangle \subseteq G'.$$

Using the previous problem, conclude that

$$\delta\Big(\Big\{Q \in \mathscr{P}_F^1 : \Big(\frac{FE'/F}{Q}\Big) = C'\Big\}\Big) = \frac{1}{f'}$$

where $f' = \mathrm{Card}(\langle C' \rangle)$.

(c) For $P \in S_K^1(C)$, define $\alpha_P(F) = \mathrm{Card}(\{Q \in \mathscr{P}_F^1 : Q \mid P\})$. Show that

$$\alpha_P(F) = \frac{1}{\mathrm{Card}(H)} \mathrm{Card}(\{\tau \in \mathcal{G} : \tau C \tau^{-1} \in H\}).$$

[*Hint*: Use [RV99, Proposition 6-2].]

(d) Put $f = \mathrm{Card}(C)$. Assuming that $f \mid f'$, show for $P \in S_K^1(C)$ that

$$\Big(\frac{L/K}{P}\Big) \cap H = \varnothing.$$

Exercise 7.20. We continue in the context of the previous problem and always assume that $f \mid f'$.

(a) Show that

$$\lim_{s \to 1^+} \sum_{P \in S_K^1(C,C')} \frac{\alpha_P(F)}{N(P)^s} \sim \frac{1}{f'} \log\Big(\frac{1}{s-1}\Big)$$

where

$$S_K^1(C,C') = \Big\{P \in S_K^1(C) : \Big(\frac{E'/K}{P}\Big) = C'\Big\}.$$

(b) Show that

$$\delta(S_K^1(C,C')) = \frac{f}{nn'}$$

where n and n' denote $\mathrm{Card}(G)$ and $\mathrm{Card}(G')$, respectively. [*Hint*: Show that $\alpha_P(F) = nn'/(ff')$.]

(c) Let

$$n'_f = \mathrm{Card}(\{\sigma' \in G' : f \mid o(\sigma')\}).$$

Show that

$$\underline{\delta}(S_K(C)) \geq \frac{fn'_f}{nn'}$$

where again $\underline{\delta}$ denotes lower Dirichlet density, as defined above. [*Hint*: Any absolutely convergent summation indexed over $P \in S_K^1(C)$ can be rewritten as a double summation, first over $\sigma' \in G'$ and then over

$$\Big\{P \in S_K^1(C) : \Big(\frac{E'/K}{P}\Big) = \sigma'\Big\}.$$

Use this and appeal to part (b).]

Exercise 7.21. We continue in the context of the previous two problems.

(a) Show that we can choose G' such that the quotient n'_f/n' is arbitrarily close to 1. [*Hint*: Suppose that f has prime factorization

$$f = \prod_{j=1}^{r} p_j^{a_j} \quad (a_j > 0).$$

Then take G' of order

$$\prod_{j=1}^{r} p_j^{b_j}$$

such that for each j, $p_j^{b_j} - p_j^{a_j - 1}$ elements of G' have order a power of $p_j^{a_j}$.]

(b) Use the previous exercise and the first part of this one to show that

$$\underline{\delta}(S_K(C)) \geq \frac{f}{n}.$$

(c) Show that

$$\underline{\delta}(S_K(G - C)) \geq \frac{n - f}{n}.$$

[*Hint*: $G - C$ is the disjoint union of conjugacy classes different from C.]

(d) Show that $\delta(S_K(C))$ exists and is given by $f/n = \mathrm{Card}(C)/\mathrm{Card}(G)$. [*Hint*: Use part (c) to conclude that

$$\overline{\delta}(S_K(C)) \leq \frac{f}{n}.$$

But the lower and upper Dirichlet densities always exist, and the latter bounds the former from above.]

Exercise 7.22. Let K be a function field in one variable over a finite field $\mathbb{F}_q$. The first parts of this exercise lead to a proof of the following result:

THEOREM. *The function $\zeta_K(s)$ is a rational function of $T = q^{-s}$. More precisely, there is a polynomial P of degree $2g$, where g is the genus of K, with $P(0) = 1$, such that*

$$\zeta_K(s) = \frac{P(T)}{(1 - T)(1 - qT)}.$$

Furthermore, $P(1) = h$, the order of the divisor class group $\mathrm{Pic}^0(K)$. (See [RV99, Section 7.5].)

REMARK. A celebrated theorem of A. Weil asserts in addition that $P(T)$ lies in $\mathbb{Z}[T]$ and admits a factorization over $\mathbb{C}$ of the form

$$\prod_{j=1}^{2g} (1 - \alpha_j T)$$

with each α_j an algebraic integer of absolute value $q^{1/2}$. Consequently, every zero of $\zeta_K(s)$ lies on the line $\{\mathrm{Re}(s) = 1/2\}$—in other words, the *Riemann hypothesis* holds for K! But a proof of this result lies far beyond the scope of this book.

(a) Show that there exist infinitely many places v of K such that $N(v) = q$. Here the norm of v simply measures the cardinality of the residue field associated with the corresponding local field K_v. [*Hint*: Suppose not. Then

$$\zeta_K(s) = \prod_{v \in S} (1 - q^{-s})^{-1} \cdot \prod_{v \notin S} (1 - q^{-f_v s})^{-1}$$

for a finite set S, with $f_v \geq 2$ for all v outside S. Deduce from this that in fact, $\log \zeta_K(s)$ has a finite limit as $s \to 1^+$, and thus derive a contradiction to [RV99, Theorem 7-19].]

(b) Show that $\zeta_K(s)$ is a meromorphic function of q^{-s}. [*Hint*: Show that the image of $|\cdot|_{\mathbb{A}_K} : \mathbb{I}_K \to \mathbb{R}_+^\times$ is $q^{\mathbb{Z}}$, and then use that $\zeta_K(s) = L(|\cdot|_{\mathbb{A}_K}^s)$.]

(c) Using part (b) and [RV99, Theorem 7-19], show that $\zeta_K(s)$ takes the form given in the theorem above, with $P(q^{-s})$ an entire function.

(d) Using the functional equation for $\zeta_K(s)$, deduce the following functional equation for the numerator:

$$P(T) = q^g T^{2g} P\Big(\frac{1}{qT}\Big).$$

Deduce that P must be a polynomial, and (by using [RV99, Theorem 7-22]) that $P(1) = h$.

(e) Let χ be any idele class character of K. Show that $L(\chi)$ is a rational function of $T = q^{-s}$. [*Hint*: Reduce to the unitary case and argue as above. Provided that $\chi \neq |\cdot|_{\mathbb{A}_K}^s$, the function $L(\chi)$ is in fact a polynomial in T.]

Exercise 7.23. Let K be a number field. For $x > 0$, put

$$\pi_K(x) = \mathrm{Card}(\{P \text{ prime in } \mathfrak{o}_K : N(P) \leq x\}).$$

The "prime number theorem" for K then asserts that as $x \to \infty$,

$$\pi_K(x) \sim \frac{x}{\log x}. \tag{7.21}$$

For $K = \mathbb{Q}$, this was conjectured by Riemann and proven independently by Hadamard and de la Vallée-Poussin in 1896. The object of this exercise is to deduce this theorem from the following theorem for Dirichlet series:

THEOREM (Tauberian Theorem for Dirichlet Series). *Let*

$$D(s) = \sum_{n \geq 1} \frac{a_n}{n^s}$$

be a Dirichlet series with nonnegative coefficients a_n that satisfies the following conditions:

(i) *$D(s)$ is normally convergent in $\{\mathrm{Re}(s) > 1\}$.*

(ii) *$D(s)$ is invertible on the line $\{\mathrm{Re}(s) = 1\}$ except for a simple pole at the point $s = 1$ of residue α.*

Then

$$\sum_{n \leq x} a_n \sim \alpha x$$

asymptotically in x.

(a) Show that

$$-\zeta_K'(s)/\zeta_K(s) = D(s) + \varphi(s)$$

for some function $\varphi(s)$ holomorphic and nonzero in $\{\mathrm{Re}(s) \geq 1\}$ and $D(s)$ the Dirichlet series defined by $a_n = \log n$, if n is the norm of some prime P, and $a_n = 0$, otherwise.

(b) Using the results of [RV99, Section 7.7], show that the function $D(s)$ of the previous part satisfies the hypotheses of the Tauberian theorem and has residue 1 at $s = 1$. Deduce that

$$g(x) = \sum_{N(P) \leq x} \log N(P) \sim x.$$

(c) Put $b_n = a_n/\log n$. Show that for any $m \geq 1$, we have

$$\pi_K(m) = \sum_{n \leq m} \frac{a_n}{\log n} = \frac{g(m)}{\log m} + h(m)$$

where

$$h(m) = \sum_{n<m} g(n)\Big[\frac{1}{\log n} - \frac{1}{\log(n+1)}\Big].$$

[*Hint*: $a_n = g(n) - g(n-1)$.]

(d) Using that $g(n) \sim n$, prove (7.21) by showing that $h(m)$ is $o(m/\log m)$; that is,

$$\lim_{m\to\infty} \frac{\log m}{m} h(m) = 0.$$

[*Hint*: Show first that

$$\Big(\frac{1}{\log n} - \frac{1}{\log(n+1)}\Big) \leq \frac{1}{n \log^2 n}$$

for all $n > 1$.]

Bibliography

[RV99] D. Ramakrishnan and R. J. Valenza, *Fourier Analysis on Number Fields*, Graduate Texts in Mathematics **186**, Springer-Verlag, New York, 1999.